



Aviation Incident Analysis Engine

Select Analysis Mode: ?



Diagnosis Mode



Prognosis Mode

Enter a description of an aviation incident or current symptom. The system will predict **what future events will happen** based on historical event sequences from similar NTSB incidents.



Example queries: 'loss of engine power during cruise flight', 'engine fire during takeoff', 'bird strike during approach'

Prognosis Method

Choose prognosis approach: ?

Event Sequence Prediction (Professor's Method)

Time-to-Failure (Legacy)

Enter Incident Description / Current Symptom:

loss of engine power during cruise flight

Predict Future Events

Prognosis Results

How Prognosis Works (Overview): Your query is converted into an embedding, compared against 2 million NTSB incidents using cosine similarity, and the top 50 most similar are retrieved. These 50 are grouped into clusters. Within each cluster, we extract the sequence of events and count what event comes after the last event. Finally, we combine all clusters using the chain rule to get one final prediction, then extend it using transition probabilities for multi-step forecasting.

Incident Clustering Analysis

What's happening here: We embedded the query and compared it against all incidents using cosine similarity to find the most similar ones. Then we grouped them by their pre-computed cluster labels (K-means, etc.). Each cluster represents a distinct incident type. $P(\text{Cluster}|\text{Query})$ tells us how relevant each cluster is. It is calculated as $(\text{avg similarity} \times \text{number of incidents}) / \text{total weight}$.

Clustered 45 incidents into 8 types

Incident Type	$P(\text{Cluster} \text{Query})$	Incidents	Sequence
Turbine Blade Failure	43.5%	19	
Insufficient Information	22.1%	10	
Engine Fire	12.5%	6	
Ice Ingestion	11.9%	5	
Loss Of Control	4.0%	2	
Bird Strike	2.2%	1	
Manufacturing Error	2.0%	1	
Electrical Failure	1.9%	1	

Future Event Prediction

Method: Conditional probability with chain rule: $P(\text{NextEvent}|\text{Query}) = \sum P(\text{NextEvent}|\text{Cluster}) \times P(\text{Cluster}|\text{Query})$

Analyzed 45 similar historical NTSB incidents

Step 1: Identify Incident Types (Clusters)

Same as the clustering table above — this shows how much weight each cluster gets in the final prediction. Higher P is more relevant to the query.

First, we calculate the probability of each incident type given the query:

$$P(Cluster \mid Query)$$

Step 2: Analyze Downstream Events within Clusters

What's happening here: For each cluster, we look at the sequence of events in each incident. We find the defining event (the anchor), then extract what comes immediately after it. We count how often each next event occurs and divide by the total to get $P(NextEvent \mid Cluster)$. This tells us: given this type of incident, what event is most likely to happen next?

For each cluster, we identify the events that follow the defining event:

$$P(NextEvent \mid Cluster)$$

 Turbine Blade Failure (P=43.5%)

Downstream events observed in Turbine Blade Failure incidents:

• Enroute-cruise - Fire/smoke (non-impact)	22.2%
• Initial climb - Engine shutdown	22.2%
• Enroute-cruise - Engine shutdown	16.7%
• Takeoff-rejected takeoff - Loss of engine power (partial)	11.1%
• Initial climb - Powerplant sys/comp malf/fail	5.6%
• Takeoff - Loss of engine power (total)	5.6%
• Enroute - Engine shutdown	5.6%
• Enroute - Loss of engine power (total)	5.6%

Sample event sequences:

 20110317X14552: Enroute-cruise - Powerplant sys/comp malf/fail → Enroute-cruise - Fire/smoke (non-impact) → Enroute-cruise - Loss of engine power (total) → Enroute-cruise - Engine shutdown → Enroute-cruise - Sys/Comp malf/fail (n

 Engine Fire (P=12.5%)
20110922X14716: Initial climb - Loss of engine power (total) → Initial climb - Powerplant sys/comp malf/fail → I
Fire/smoke (non-impact)

Downstream events observed in Engine Fire incidents:

• Enroute-climb to cruise - Engine shutdown	50.0%
• Initial climb - Engine shutdown	50.0%

Sample event sequences:

 20110105X33913: Enroute-climb to cruise - Fire/smoke (non-impact) → Enroute-climb to cruise - Powerplant s
malf/fail → Enroute-climb to cruise - Engine shutdown

 20110105X33913: Enroute-climb to cruise - Fire/smoke (non-impact) → Enroute-climb to cruise - Powerplant s
malf/fail → Enroute-climb to cruise - Engine shutdown

 Ice Ingestion (P=11.9%)

Downstream events observed in Ice Ingestion incidents:

- Initial climb - Powerplant sys/comp malf/fail 100.0%

Sample event sequences:

20140714X45633: Initial climb - Fire/smoke (non-impact) → Initial climb - Powerplant sys/comp malf/fail → Init Engine shutdown → Initial climb - Loss of engine power (total) → Initial climb - Electrical system malf/failure

20140714X45633: Initial climb - Fire/smoke (non-impact) → Initial climb - Powerplant sys/comp malf/fail → Init Engine shutdown → Initial climb - Loss of engine power (total) → Initial climb - Electrical system malf/failure

Loss Of Control (P=4.0%)

Downstream events observed in Loss Of Control incidents:

- Uncontrolled descent - Flight control sys malf/fail 100.0%

Sample event sequences:

20130429X12734: Prior to flight - Aircraft loading event → Initial climb - Loss of control in flight → Uncontrolled control sys malf/fail

Step 3: Final Event Prognosis

What's happening here: We combine results from ALL clusters using the chain rule. For each possible event sequence we multiply $P(\text{NextEvent}|\text{Cluster}) \times P(\text{Cluster}|\text{Query})$ for every cluster, then sum them up. This gives us the probability that weighs each cluster's contribution by how relevant it is to the query. The 'Calculation' column shows exactly which clusters contributed and how much.

Combined probability using the Total Probability Theorem:

$$P(E | Q) = \sum P(E | I) \times P(I | Q)$$

1. Initial climb - Engine shutdown — Probability: 15.9%

Probability: 15.9%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.2222 \times 0.4348) + (0.5000 \times 0.1250) = \mathbf{0.1591 (15.9\%)}$

Contributing clusters: 2

Cluster-by-cluster breakdown:

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.2222 \times P(\text{Cluster}|\text{Q})=0.4348 = 0.0966$

Engine Fire: $P(\text{Event}|\text{Cluster})=0.5000 \times P(\text{Cluster}|\text{Q})=0.1250 = 0.0625$

Supporting incidents: 20170608X90705, 20020917X01767, 20140922X14716

2. Initial climb - Powerplant sys/comp malf/fail — Probability: 14.3%

Probability: 14.3%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (1.0000 \times 0.1186) + (0.0556 \times 0.4348) = \mathbf{0.1427 (14.3\%)}$

Contributing clusters: 2

3. Enroute-cruise - Fire/smoke (non-impact) — Probability: 9.7%

Cluster-by-cluster breakdown:

Ice Ingestion: $P(\text{Event}|\text{Cluster})=1.0000 \times P(\text{Cluster}|\text{Q})=0.1186 = 0.1186$

Probability: 9.7%

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.0556 \times P(\text{Cluster}|\text{Q})=0.4348 = 0.0242$

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.2222 \times 0.4348) = \mathbf{0.0966 (9.7\%)}$

Supporting incidents: 20170608X90705, 20020612X00866, 20140922X14716

Contributing clusters: 1

Cluster-by-cluster breakdown:

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.2222 \times P(\text{Cluster}|\text{Q})=0.4348 = 0.0966$

Supporting incidents: 20170608X90705, 20140922X14716, 20120221X03231

4. Enroute-cruise - Engine shutdown — Probability: 7.2%

Probability: 7.2%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.1667 \times 0.4348) = \mathbf{0.0725 (7.2\%)}$

Contributing clusters: 1

Cluster-by-cluster breakdown:

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.1667 \times P(\text{Cluster}|\text{Q})=0.4348 = 0.0725$

Supporting incidents: 20170608X90705, 20140922X14716, 20120221X03231

5. Enroute-climb to cruise - Engine shutdown — Probability: 6.2%

Probability: 6.2%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.5000 \times 0.1250) = \mathbf{0.0625 (6.2\%)}$

Contributing clusters: 1

Cluster-by-cluster breakdown:

Engine Fire: $P(\text{Event}|\text{Cluster})=0.5000 \times P(\text{Cluster}|\text{Q})=0.1250 = 0.0625$

Supporting incidents: 20020917X01767, 20001213X27987, 20160707X02944
6. Takeoff-rejected takeoff - Loss of engine power (partial) — Probability: 4.8%

Probability: 4.8%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.1111 \times 0.4348) = \mathbf{0.0483 (4.8\%)}$

Contributing clusters: 1

Cluster-by-cluster breakdown:

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.1111 \times P(\text{Cluster}|\text{Q})=0.4348 = 0.0483$

Supporting incidents: 20170608X90705, 20140922X14716, 20120221X03231

7. Uncontrolled descent - Flight control sys malf/fail — Probability: 4.0%

Probability: 4.0%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (1.0000 \times 0.0399) = \mathbf{0.0399 (4.0\%)}$

Contributing clusters: 1

Cluster-by-cluster breakdown:
8. Enroute - Loss of engine power (total) — Probability: 2.4%

Loss of Control: $P(\text{Event}|\text{Cluster})=1.0000 \times P(\text{Cluster}|\text{Q})=0.0399 = 0.0399$

Probability: 2.4%
Supporting incidents: 20110401X24330, 20130429X12734

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0556 \times 0.4348) = \mathbf{0.0242 (2.4\%)}$

Contributing clusters: 1

Cluster-by-cluster breakdown:

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.0556 \times P(\text{Cluster}|\text{Q})=0.4348 = 0.0242$

Supporting incidents: 20170608X90705, 20140922X14716, 20120221X03231

9. Enroute - Engine shutdown — Probability: 2.4%

Probability: 2.4%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0556 \times 0.4348) = \mathbf{0.0242 (2.4\%)}$

Contributing clusters: 1

Cluster-by-cluster breakdown:

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.0556 \times P(\text{Cluster}|\text{Q})=0.4348 = 0.0242$

Supporting incidents: 20170608X90705, 20140922X14716, 20120221X03231

10. Enroute - Loss of engine power (total) — Probability: 2.4%

Probability: 2.4%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0556 \times 0.4348) = \mathbf{0.0242 (2.4\%)}$

Contributing clusters: 1

Cluster-by-cluster breakdown:

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.0556 \times P(\text{Cluster}|\text{Q})=0.4348 = 0.0242$

Supporting incidents: 20170608X90705, 20140922X14716, 20120221X03231

Multi-Step Event Sequence (Bayesian Chain

What's happening here: Step 1 predicts the immediate next event. But incidents cascade — $A \rightarrow B \rightarrow$ Step 2 and Step 3, we use transition probabilities built from consecutive event pairs in the retrieved sequences. For each Step 1 event, we look at what historically follows it, then weight by how likely t event was. This chains the prediction forward through multiple steps.

Extending the prediction forward using transition probabilities:

$$P(Event_t \mid Q) = \sum P(Event_t \mid Event_{t-1}) \times P(Event_{t-1} \mid Q)$$

Step 1: Immediate Next Event

These are the same results from Step 3 above — the most likely events to happen immediately after the query sympt

Initial climb - Engine shutdown	P
	15
Initial climb - Powerplant sys/comp malf/fail	P

Enroute-cruise - Fire/smoke (non-impact)	P	14
		9.7
Enroute-cruise - Engine shutdown	P	
		7.2
Enroute-climb to cruise - Engine shutdown	P	
		6.2

◆ Step 2: What Follows After

These events are predicted by looking at what historically follows Step 1 events in the sequence data, weighted by e: probability.

Initial climb - Fire/smoke (non-impact)	P	14
Enroute-cruise - Loss of engine power (total)	P	4.2
Enroute-cruise - Sys/Comp malf/fail (non-power)	P	3.2
Initial climb - Engine shutdown	P	2.7
Initial climb - Loss of engine power (total)	P	1.9

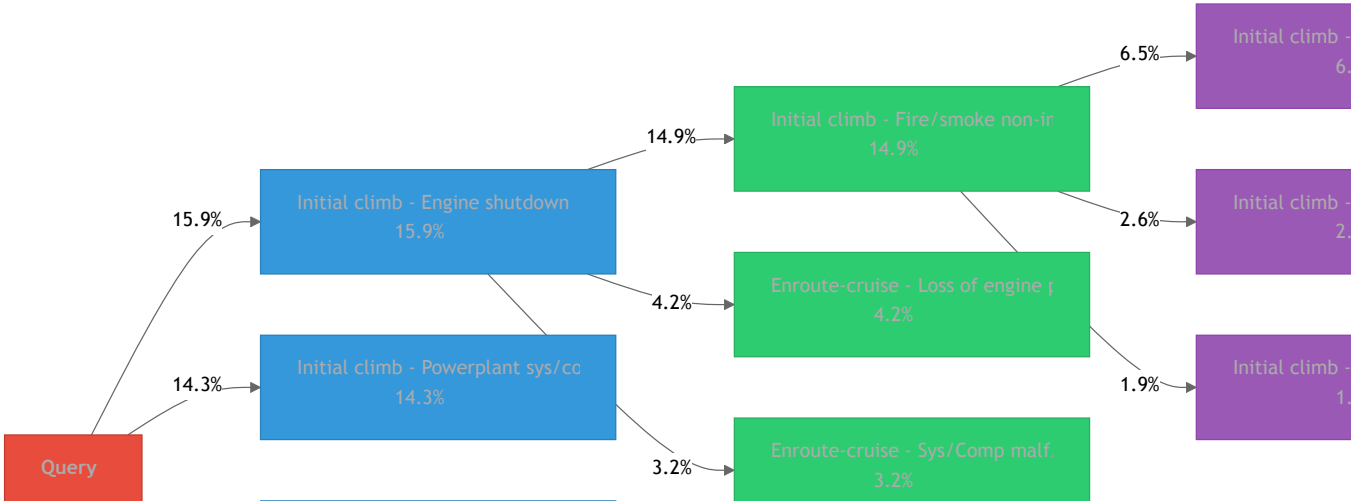
◆ Step 3: What Follows After

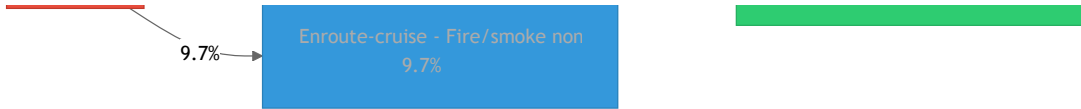
These events are predicted by looking at what historically follows Step 2 events in the sequence data, weighted by event probability.

Initial climb - Cabin safety event	P	6.5%
Initial climb - Powerplant sys/comp malf/fail	P	2.6%
Initial climb - Engine shutdown	P	1.9%
Enroute-cruise - Engine shutdown	P	1.8%
Initial climb - Fire/smoke (non-impact)	P	1.2%

 Predicted Event Cascade

This diagram shows the most likely chains of events from the query symptom. Each arrow shows the transition probability between events. Read left to right: Query → Step 1 → Step 2 → Step 3.





Similar Historical Incidents

Top 4 most similar incidents

Incident: 20081201X44308 (Similarity: 80.62%)

Incident: 20110317X14552 (Similarity: 80.31%)

Incident: 20170607X22245 (Similarity: 80.31%)

Incident: 20140714X45633 (Similarity: 71.14%)